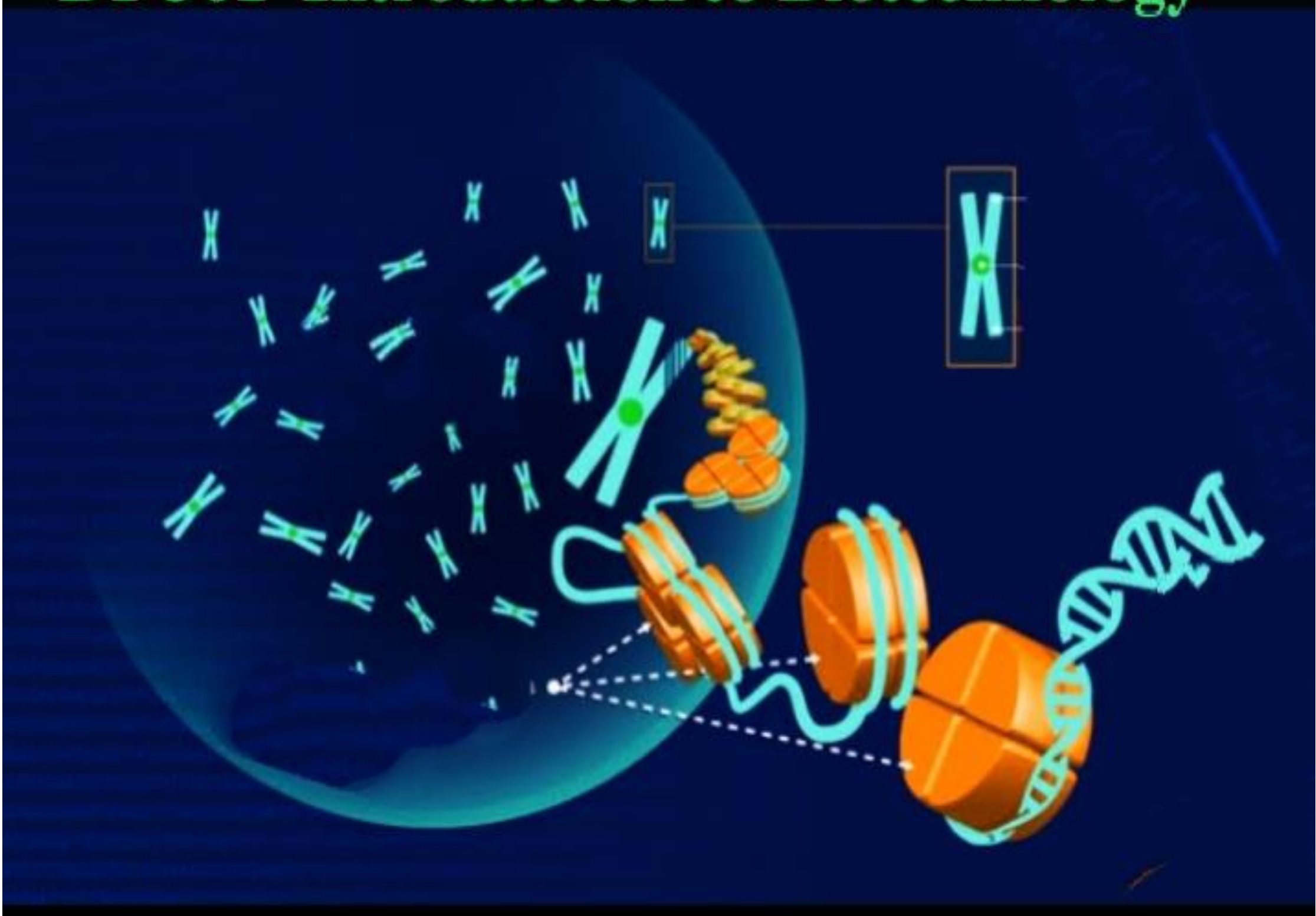
Assignment No.1

BT-301 Introduction to Biotechnology



By

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Solution:

Question Number.1

Elucidation of DNA structure:

DNA is a polymer composed of two polynucleotide chains that coil around each other to form a double helix. The polymer carries genetic instructions for the development, functioning, growth and reproduction of all known organisms and many viruses.

Taken in 1952, this image is the first X-ray picture of DNA, which led to the discovery of its molecular structure by Watson and Crick. Created by Rosalind Franklin using a technique called X-ray crystallography, it revealed the helical shape of the DNA molecule. Watson and Crick realized that DNA was made up of two chains of nucleotide pairs that encode the genetic information for all living things.

By the early 1950s, at least two groups were hot on the trail. Crick, a British graduate student, and Watson, an American research fellow, were in the hunt at Cambridge University.

At King's College in London, Rosalind Franklin and Maurice Wilkins were studying DNA. Wilkins and Franklin used X-ray diffraction as their main tool -- beaming X-rays through the molecule yielded a shadow picture of the molecule's structure, by how the X-rays bounced off its component parts.

Franklin, a shy and inward young woman, suffered from patronizing attitudes and sexism that forced her to do much of her work alone. And her senior partner, Wilkins, showed some of Franklin's findings to Watson in January 1953 without her knowledge.

Referring to Franklin's X-ray image known as "Exposure 51," James Watson is reported to have said, "The instant I saw the picture, my mouth fell open and my pulse began to race." Shortly after, Watson and Crick made a crucial advance when they proposed that the DNA molecule was made up of two chains of nucleotides paired in such a way to form a double helix, like a spiral staircase. This structure, announced in their famous paper in the April 1953 issue of Nature, explained how the DNA molecule could replicate itself during cell division, enabling organisms to reproduce themselves with amazing accuracy except for occasional mutations.

Importance of DNA structure:

The units (monomers) that make a DNA molecule are called nucleotides. A single nucleotide is comprised of a deoxyribose sugar, phosphate molecule, and a nitrogenous base. There are four nitrogenous bases in DNA: adenine, thymine, guanine, and cytosine.

Knowing the structure of DNA is important because DNA is the molecule that contains the genetic information required for a cell to make proteins. Proteins are required for all cellular activities to take place and a cell would not be able to carry out any functions necessary to remain a living unit without proteins. Having knowledge of the structure of DNA is critical since proteins are solely formed using the information stored in the structure of DNA.

DNA's unique structure enables the molecule to copy itself during cell division. When a cell prepares to divide, the DNA helix splits down the middle and becomes two single strands. These single strands serve as templates for building two new, double-stranded DNA molecules - each a replica of the original DNA molecule.

Question Number.2

Structure-Function relationship of Proteins:

Protein structure sets the foundation for its interaction with other molecules, which ultimately determines its function. Proteins fold up into specific shapes according to the sequence of amino acids in the polypeptide chain. If the specific 3D structure is disrupted, for which the protein is said to be denatured, the protein loses its activity. For example, denatured enzymes lose their catalytic ability, denatured antibodies can no longer bind antigen, and disrupted subunits cannot assemble into multimers to perform their functions. In a word, the protein function is directly dependent on the protein structure, which cannot be carried out once the structure is disturbed.

Understanding the relationships between protein structure and function remains a primary focus in structural biology with important consequences in such diverse fields as molecular biology, genetics, biochemistry, protein engineering and bioinformatics. One approach to this problem is to study how nature has re-engineered proteins for new functions through evolutionary processes. This strategy has the potential to reveal fundamental characteristics of protein structures and the explicit manner in which they deliver their associated functions.

To understand the structure-function paradigm, particularly useful structural information comes from the primary amino acid sequences and the associated tertiary structures. Several recent developments in analysis of the "protein universe" at the tertiary structural level have provided important criteria for understanding the range of family folds that exist and some of the evolutionary relationships associated with them. While the tertiary structure database is small, the sequence databases are large and now include the sequences of the entire genomes of several bacteria, an archaeon and a microbial eukaryote. Many additional genomes will be solved in the near future. Using the great deal of protein sequence and structural data at hand, computational strategies to address the "structure-function" problem can now support serious attempts to understand the fundamental relationships between protein structure and function.

Example:

As a result of the diverse nature of roles proteins carry out, how the polypeptide chains are joined relates specifically to its function. For example, fibrous proteins, which carry out a structural role, have a different quaternary structure to globular proteins such as enzymes that have a metabolic function.

Regards:

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